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## CURRICULUM VITÆ

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Ny Hasina Marolahy VAGNO  
Gsm : +262693428016  
Email : nyhasinavagno@gmail.com  
Website : <http://www.nyhasinavagno.fr>



### Education

2025-2027: **Engineering Degree in Computer Science** at ESIROI (final year)  
<https://esiroi.univ-reunion.fr>  
2021-2024 : **Bachelor's Degree in Software Ecosystems** at HEI  
<https://hei.school>

### Technical Skills

**Programming** : Python, Java, TypeScript  
**Web** : Spring Boot, API REST, Hibernate, JUnit, Testcontainers, Next.js  
**DevOps** : AWS (Lambda, SQS, EventBridge), Docker, CI/CD  
**AI / ML** : PyTorch, Keras, YOLO, OpenCV, Scikit-learn, Pandas, GeoPandas, Seaborn, Matplotlib  
**Data** : PostgreSQL, MySQL  
**Tools** : Git, Jira, SonarQube, Kaggle, Roboflow

### Areas of Interest

- Machine learning and deep learning
- Computer vision and geographic information systems
- Data processing and feature engineering
- Software architecture, design, and quality assurance

### Professional Experience

#### Research Intern, GRC (UPV), Spain (Apr. 2026 – Aug. 2026):

- *Fine-tuning* YOLOv11 models (fruit detection, instance segmentation of tree canopies)
- Creation and annotation of a 550-image UAV dataset (6,872 trees, 80,446 fruits) assisted by SAM
- Hyperparameter optimization using *Ray Tune*; data augmentation (*crop*, *mosaic*, *contrast*)
- Image tiling inference (*SAHI*) for small-object detection at full resolution
- Ablation study on the impact of orthomosaic resampling (mAP@50: 0.97 → 0.92)
- Direct georeferencing of detections from *Structure-from-Motion* outputs (OpenDroneMap)
- Multi-view spatial deduplication (geometric NMS, STR-tree index, *Intersection over Minimum*)
- Fisher–Jenks clustering with automated *k* selection (*Goodness of Variance Fit*, silhouette)
- Model validation, prevention of *data leakage*, and evaluation on an independent test set

#### MLOps Developer, Numer, Madagascar (2022–2025):

- Deployment of a YOLOv11 model on AWS Lambda using Docker containers
- Design of an asynchronous event-driven processing architecture (SQS, EventBridge)
- Implementation of the training pipeline: data augmentation, experiment tracking
- Model validation and prevention of *data leakage* during dataset construction
- Geospatial post-processing of detections: geometrization, continuity analysis, GeoPackage export
- Participation in Agile (Scrum) ceremonies and code reviews

#### Teaching Assistant, HEI, Madagascar (2024–2025):

- Object-oriented programming (PROG2: encapsulation, inheritance, polymorphism, abstraction)
- Backend API development (PROG3: MVC, unit testing, and integration testing)
- Mathematics for Computer Science, including IEEE 754 floating-point representation
- Diffie–Hellman key exchange algorithm

### Languages

- Malagasy (native)
- French (bilingual)
- English (TOEIC – B2 level)